From China to Brazil: Two Paths, One Wisdom

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More than 40 years ago, Chinese soil scientist Guangjiong Hou proposed natural no-till farming. Around the same period, Swiss geneticist Ernst Götsch began practicing syntropic agroforestry in Brazil. Working on different lands, they arrived at the same wisdom: cooperate with nature and allow the land to restore itself.

No-till farming is like building a reservoir inside the soil: it saves water and stabilizes yields. Syntropic agroforestry is like bringing the forest into the field: pruning, mulching and layered planting help microorganisms and crops thrive together. Across regions and climates, the key is the same - reduce tillage, increase cover and raise diversity. The real path to high productivity is not to fight nature, but to learn how to work with it.

Why this case matters: Climate change, land degradation and rising production costs are placing farmers under growing pressure. How can farmers reduce external inputs while improving soil, stabilizing yields and increasing income?

Two Roads, One Direction


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More than four decades ago, Academician Guangjiong Hou proposed natural no-till farming in China. At nearly the same time, Ernst Götsch began exploring syntropic agroforestry on degraded pastureland in Brazil. Although the two approaches emerged from different soils and climates, they moved toward the same destination: working with nature to restore the vitality of the land.

Guangjiong Hou's Natural No-Till: Bringing Soil Back to Life

Hou believed that soil is not an inert material but a living ecosystem. Conventional tillage may appear to loosen soil, but it breaks soil structure, accelerates the decomposition of organic matter and makes the land less alive over time.

His natural no-till method was not a matter of doing nothing. It was a complete system of cooperation with nature, built on what he called the small three transformations and the large three transformations.

The small three transformations: improving the inside of the soil

- Structuring: apply organic fertilizers and straw mulch so the soil becomes soft and crumbly.
- Humification: increase organic matter so the dark topsoil layer gradually thickens.
- Bacterization: restore microbial communities - the invisible cattle and sheep of the soil - so they can transform residues into nutrients.

The large three transformations: improving the field environment

- Shelterbelt networks: plant windbreaks around fields to improve the microclimate and reduce wind erosion.
- Water-channel networks: build field water systems so the land can store water like a reservoir.
- Multiple cropping: use rotation and intercropping to keep the soil covered year-round and keep living roots in the ground.

In experiments in Sichuan and Guizhou during the 1980s and 1990s, no-till rice fields produced several dozen jin more per mu than conventional fields, and in some cases more than 100 jin. Water efficiency was even more striking: studies showed that no-till rice fields used water 20% to 30% more efficiently than conventional systems.

This reflects the reservoir effect of no-till soil: ridges and furrows allow rainwater to infiltrate rather than run off; cover reduces evaporation; and higher organic matter improves water-holding capacity. No-till therefore saves water while making yields more stable. After several years, as microbial abundance and diversity recover, soil fertility and resilience improve significantly.

Syntropic Cooperation: Bringing Forest Wisdom into Farmland

Original syntropic farming image preserved.

Ernst Götsch's syntropic agroforestry also emphasizes cooperation with nature. Its core idea is simple: make farmland function like a forest.

Forests recover fertility and productivity after disturbance without chemical fertilizers or pesticides because they rely on several principles: layered coexistence, dynamic succession, nutrient cycling and self-regulation.

How the system works

- Layered, high-density planting: trees, fruit trees, grain crops and groundcover plants form a three-dimensional field. Sunlight is used more fully, and photosynthetic efficiency can be 30% to 50% higher than in monoculture.
- Pruning and mulching: regular pruning stimulates growth, returns branches and leaves to the soil, adds organic matter and feeds soil microorganisms.
- Dynamic succession design: pioneer species such as nitrogen-fixing trees and fast-growing herbs restore the soil first. Vegetables, fruits and other cash crops are introduced gradually, eventually forming a stable perennial system.

- Water management: contour planting, deep-rooted plants and rainwater harvesting reduce soil erosion and water loss. In a Brazilian case, six streams that had run dry began flowing again.
- Ecological self-regulation: diverse crops and rich microbial communities create a natural protection network, greatly reducing the need for external fertilizers and pesticides.

Under these practices, degraded land continues to recover. Experiments in Peru showed a visible thickening of the dark topsoil layer. Research in Belgium found that agroforestry fields stored about one quarter more carbon than monoculture fields. In Brazil, the land even helped revive dried streams. Götsch's practice demonstrates that when farmers cooperate with nature, degraded land can become green and productive again.

China's No-Till and Brazil's Syntropy: Different Paths, Shared Wisdom

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Hou's natural no-till approach emphasizes the small three and large three transformations, while Götsch's syntropic agroforestry emphasizes forest principles. Their starting points differ, yet their conclusion is the same: sustainable agriculture depends on cooperation with nature.

At the level of soil improvement, no-till uses organic fertilizer and straw to improve structure, increase organic matter and restore microbes. Götsch achieves similar effects through layered roots, pruned biomass and diverse vegetation. At the field level, no-till emphasizes shelterbelts, water networks and multiple cropping; syntropic systems use mixed tree planting, contour lines, deep-rooted species and dense multi-species combinations to keep the land covered all year.

Their shared lesson is this: reducing tillage and external inputs does not mean conquering the land. It means turning ecological processes into the driving force of agriculture.

Local Challenges and Possibilities

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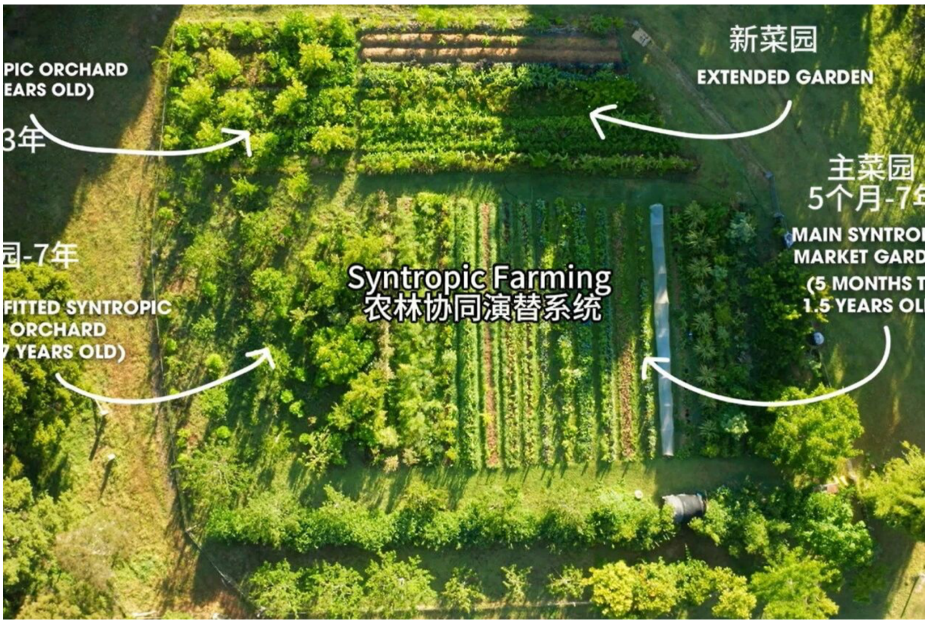
Many farmers worry that no-till will be difficult to sustain, or that Götsch's method only works in the tropics. In reality, the question is not where a model comes from, but how its principles can be localized. The challenge is to find local ways of cooperating with nature.

No-till is not passive neglect. It replaces tillage with cover, microorganisms and crop combinations. At first, yields may fluctuate, but after several years soil recovery increases, production becomes more stable and labor can decline. The system also becomes more resilient under extreme weather.

Götsch's method is similar. Its core is layering, diversity and cycling. These principles can work anywhere. The real challenge is choosing crop combinations suited to local conditions. As long as farmers increase diversity, reduce tillage, plant in layers, prune wisely and add cover, they can design their own way to work with nature.

Looking Ahead: New Opportunities for China's Smallholders



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Although natural no-till did not become mainstream when it was first proposed, its ideas remain forward-looking today. Under China's dual-carbon goals, agriculture must reduce emissions and increase carbon sinks. No-till and syntropic agroforestry offer practical paths.

- Carbon storage: soil can become a major carbon reservoir.
- Higher and more stable productivity: restored microbial diversity helps crops resist drought and disease.
- Ecological restoration: diversified planting improves microclimates and increases biodiversity.

Conclusion: Working with Nature

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From the rice fields of China to the forests of Brazil, two approaches emerged in the same era and from different landscapes, yet reached the same answer: the future of agriculture does not lie in fighting nature. It lies in learning to cooperate with it.

For small farmers, even one less tillage pass, a little more cover or one additional companion crop can be a step toward cooperation with nature. As soil microorganisms return, the land becomes more fertile and life becomes more secure.

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About the Soil Vitality Lab

The Soil Vitality Research and Enhancement Laboratory at Sichuan University was established in 2021. Its core members include Dr. Hongyan Lu, Dr. Jiong Yan, Dr. Xue Chen and Ms. Wei Tang. The lab promotes dialogue between field practice and science, supporting farmers as they explore localized pathways to rebuild soil vitality and strengthen agricultural climate resilience.